Monte Carlo Proxy Pricing

A step-by-step methodology for European, American, Asian, barrier, cliquet, ten-asset basket Asian, SLV cliquet, and basket cliquet instruments

Written for a second-year undergraduate with calculus, probability, and linear algebra

This guide derives the transformations, estimators, regression bases, interpolants, variance-reduction methods, dynamic-programming equations, and validation metrics used in the proxy-pricing experiments.

Instrument	Effective state	Current preferred fitter
European call	1D d1-like moneyness	PCHIP default; Bernstein optimized
American put	Spot and exercise time	PCHIP continuation recursion
Arithmetic Asian	1D adjusted moneyness after reduction	PCHIP default; Akima optimized
10-asset basket Asian	10 spots plus running basket sum	Moment baseline + PCA sparse-Chebyshev correction
Barrier call	Spot plus alive/hit flag	PCHIP + Brownian bridge
GBM cliquet	1D accrued clipped return	Bounded Chebyshev degree 19
Single-name SLV cliquet	Accrued, spot, variance	Local/spectral hybrid
3-asset SLV basket cliquet	Accrued, 3 spots, 3 variances	Sparse baseline; enrichment needed

Research implementation: proxy_pricing repository

1. Scope and guiding principle

A pricing proxy is a fast approximation V_hat(x,t) to an expensive risk-neutral value V(x,t). The central difficulty is not regression alone. It is choosing a Markov state, transforming the payoff geometry, generating unbiased low-noise labels, fitting without violating known bounds, and testing in the tails.

The experiments support one broad rule: use product structure to reduce dimension first, then choose the simplest smoother compatible with the remaining geometry. A one-dimensional curve and a seven-dimensional SLV surface should not be forced into the same numerical representation.

How to read this guide

The main chapters explain the ideas operationally: what is being priced, what state is needed, how Monte Carlo labels are generated, and how the proxy is fitted. Proofs are postponed to the appendices so that the pricing workflow remains visible. Every theorem used by the implementation is restated there with assumptions, a proof or proof sketch, and its role in the code.

Appendix H is a mathematical audit of the implementation tricks. It states the exact claim behind state enrichment, Sobol batching, PCA compression, moment-matched basket baselines, ridge residual fitting, log-factor targets, PCHIP residual calibration, relative-error floors, and exact payoff wings.

Minimal prerequisites and notation

Symbol	Meaning
S_t	Underlying asset price at time t
r, q, sigma	Interest rate, dividend yield, volatility
E[. | X_t=x]	Conditional expected value given today's state
V(t,x)	True model price; V_hat(t,x) is its proxy
N	Number of Monte Carlo paths
SE	Estimated Monte Carlo standard error
clip(u,L,U)	min(max(u,L),U)

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What the reported error means

Unless stated otherwise, relative error is abs(proxy-benchmark) divided by max(abs(benchmark), 0.01). The 0.01 floor prevents an economically tiny absolute discrepancy near zero from dominating a percentage statistic. Signed and absolute errors are retained separately.

2. Risk-neutral valuation foundation

2.1 Conditional expectation

Under a risk-neutral measure Q, with short rate r and state X_t, a European-style discounted payoff H has value

```
V(t,x) = E_Q[ exp(-Integral_t^T r_u du) H(X_T) | X_t = x ].
```

Path-dependent contracts become Markov after augmenting X_t with sufficient path statistics: a running sum for an Asian, accrued clipped return for a cliquet, and current variances for SLV.

2.2 GBM transition

For constant parameters under Q:

```
dS_t / S_t = (r-q) dt + sigma dW_t.

Integrating d log(S_t) gives
log(S_T/S_t) = (r-q-sigma^2/2) tau + sigma sqrt(tau) Z,
Z ~ N(0,1).

Therefore
S_T = S_t exp((r-q-sigma^2/2) tau + sigma sqrt(tau) Z).
```

This exact transition removes time-discretization error for GBM European, Asian-fixing, and GBM cliquet simulations.

2.3 SLV transition

```
dS_i/S_i = (r-q_i) dt + L_i(S_i) sqrt(v_i) dW_i^S
dv_i     = kappa_i(theta_i-v_i) dt + xi_i sqrt(v_i) dW_i^v.
```

The experiments use full-truncation Euler. With v_plus=max(v,0):

```
S_next = S exp((r-q-0.5 L(S)^2 v_plus) dt
               + L(S) sqrt(v_plus dt) Z_S)

v_next = max(v + kappa(theta-v_plus) dt
               + xi sqrt(v_plus dt) Z_v, 0).
```

This preserves positivity numerically but introduces discretization bias. Proxy error is measured against a benchmark using the same grid, so Euler bias is outside the reported proxy metric.

2.4 Universal pricing-label workflow

For every instrument, the expensive training label is constructed before any proxy is fitted. At state x_i and valuation date t, the reusable sequence is:

Step	Pricing action	Output
1	Reconstruct the sufficient Markov state	x_i
2	Simulate risk-neutral transitions or paths	X^(j)
3	Evaluate path payoff and discount	Y_i^(j)
4	Apply unbiased controls or likelihood ratios	Y_tilde_i^(j)
5	Average paths and estimate sampling error	V_i and SE_i
6	Repeat on independent validation states	Benchmark surface

```
Training label:
V_i = (1/N_i) Sum_(j=1)^N_i Y_tilde_i^(j)

Estimated label variance:
Var(V_i) = sample_var(Y_tilde_i) / N_i.
```

A proxy approximates the conditional expectation represented by these labels; it does not replace the risk-neutral pricing definition. Exact payoff regions are evaluated analytically and removed from the regression whenever possible.

3. Monte Carlo labels and variance reduction

3.1 Plain estimator and uncertainty

```
Y_j = discounted payoff on path j
V_hat_MC = (1/N) Sum_j Y_j
SE(V_hat_MC) = sample_std(Y) / sqrt(N).
```

The standard error falls only as N^(-1/2). Ten times smaller noise requires one hundred times as many paths, so structural variance reduction is more valuable than path count alone.

3.2 Antithetic sampling

```
For Z ~ N(0,I), both Z and -Z have the same law.
Use pair average: Y_pair = (Y(Z)+Y(-Z))/2.

Var(Y_pair) = 0.5 Var(Y) + 0.5 Cov(Y(Z),Y(-Z)).
```

For monotone payoffs the covariance is often negative, reducing variance without changing the expectation.

3.3 Likelihood-ratio importance sampling

Suppose the standard normal is sampled from q=N(mu,I) instead of p=N(0,I). The density ratio follows directly by completing the square:

```
p(z) / q(z)
= exp(-z'z/2 + (z-mu)'(z-mu)/2)
= exp(-mu'z + mu'mu/2).

E_p[H(Z)] = E_q[ H(Z) p(Z)/q(Z) ].
```

For multiple independent steps, log likelihoods add. The European and standalone one-dimensional Asian defaults use shifted terminal or path normals where appropriate. SLV cliquets use a defensive mixture: half unshifted and half shifted paths.

```
q_mix(z) = 0.5 p(z) + 0.5 p(z-mu)

p(z)/q_mix(z)
= 1 / [0.5 + 0.5 exp(mu'z - mu'mu/2)].
```

The unshifted component protects central states; the shifted component stabilizes rare positive payoffs near a global floor.

The ten-asset basket Asian experiment uses both ideas. Its state design is enriched across basket level, running sum, and PCA directions. Its path generator also uses a true two-component Gaussian likelihood-ratio mixture for the long-dated OTM slice where the shift reduced variance. Later fixing dates use plain Sobol because the same shift increased weighted variance.

4. Control variates

Let Y be the target payoff and C a correlated control with known expectation c. Define Y_beta=Y-beta(C-c). It remains unbiased because E[C-c]=0.

```
Var(Y_beta) = Var(Y) + beta^2 Var(C) - 2 beta Cov(Y,C).

Differentiate with respect to beta:
2 beta Var(C) - 2 Cov(Y,C) = 0.

beta_star = Cov(Y,C) / Var(C).
```

The Asian benchmark uses the exact discrete geometric Asian value as the control expectation. The GBM cliquet uses the future clipped-return sum, whose first moment is available from truncated lognormal moments.

4.1 Truncated lognormal moment identity

If X~N(mu,s^2), then for any real a and threshold b:

```
E[ exp(aX) 1_{X <= b} ]
= exp(a mu + 0.5 a^2 s^2)
  Phi((b-mu-a s^2)/s).
```

Derivation: multiply the normal density by exp(aX), complete the square, and recognize a shifted normal density. Differences of this expression over two thresholds give interval moments. These moments produce exact means and variances for clip(exp(X)-1, local_floor, local_cap).

4.2 Benchmark independence

Training and benchmark random streams are independent. A proxy should not be evaluated against the same paths used to train it; common random numbers may make a poor proxy appear artificially accurate.

5. Proxy targets and regularized bases

5.1 Coordinate scaling

```
x_scaled = 2 (x-x_min)/(x_max-x_min) - 1.
```

Chebyshev and Bernstein systems are evaluated on compact intervals. Inputs outside the trained domain are clipped in these experiments; production systems should define explicit extrapolation policies.

5.2 Ridge regression

```
min_beta ||A beta - y||_2^2 + lambda ||Gamma beta||_2^2

Normal equation:
(A'A + lambda Gamma'Gamma) beta = A'y.
```

The intercept is not penalized. Ridge stabilizes noisy Monte Carlo labels and ill-conditioned high-order bases.

5.3 Chebyshev basis

```
T_0(x)=1, T_1(x)=x,
T_(n+1)(x)=2x T_n(x)-T_(n-1)(x).

Proxy: y_hat(x)=Sum_(k=0)^d beta_k T_k(x).
```

Chebyshev polynomials are better conditioned on [-1,1] than monomials and work well for globally smooth value functions.

5.3.1 Tensor basis and sparse multi-index set

For scaled state z=(z_1,...,z_D), a multivariate tensor Chebyshev term is

```
Psi_alpha(z) = Product_(j=1)^D T_(alpha_j)(z_j),
alpha = (alpha_1,...,alpha_D), alpha_j >= 0.

Full tensor degree p: 0 <= alpha_j <= p
Number of terms: (p+1)^D.
```

The full tensor is impossible in seven dimensions. Sparse regression keeps only a payoff-aware index set. A standard anisotropic form is

```
I_p = { alpha : Sum_j w_j alpha_j <= p },

where small w_j permits higher degree in an important coordinate and large w_j
suppresses degree in a weak coordinate.
```

5.3.2 Basket cliquet sparse basis

The basket implementation gives the lower and upper payoff cushions the largest degree because they locate the global floor and cap transitions. The six spot/variance coordinates enter through lower-order interactions:

```
1. T_a(lower cushion),              a=0,...,15
2. T_b(upper cushion),              b=1,...,8
3. T_a(lower) T_b(upper),           a+b<9
4. T_a(lower) state_j,              a=0,...,5
5. T_b(upper) state_j,              b=0,...,3
6. state_j state_k,                 j<k.
```

This is not a generic total-degree polynomial. It encodes the belief that payoff-boundary location is strongly nonlinear, while residual SLV state dependence is smoother and lower order.

5.3.3 Sparse Chebyshev training algorithm

```
Input: states x_i, MC values V_i, known bounds L<U.

1. Scale each feature to z_ij in [-1,1].
2. Transform p_i=clip((V_i-L)/(U-L),eps,1-eps).
3. Set y_i=log(p_i/(1-p_i)).
4. Build A_(i,k)=Psi_(alpha_k)(z_i) for selected terms.
5. Solve (A'A + lambda Gamma) beta = A'y.
6. At query x, evaluate y_hat=Psi(x)' beta.
7. Return V_hat=L+(U-L)/(1+exp(-y_hat)).
8. Override exact floor, cap, or maturity states.
```

With M retained terms and N labels, evaluation is O(M) per state. Normal-equation training is approximately O(N M^2 + M^3), but it is convex, deterministic, and normally much cheaper than path generation.

5.4 Bernstein and global Bezier basis

```
B_(k,n)(u) = C(n,k) u^k (1-u)^(n-k),  0<=u<=1.
y_hat(u) = Sum_(k=0)^n c_k B_(k,n)(u).
```

The Bernstein basis is nonnegative and sums to one. With fitted coefficients c_k it is a global Bezier curve. It provided the best European 1D result.

5.5 Log and bounded-logit targets

```
Positive value: y = log(V + epsilon), V_hat=exp(y_hat)-epsilon.

Known bounds L<U:
p=(V-L)/(U-L), y=log(p/(1-p)),
V_hat=L+(U-L)/(1+exp(-y_hat)).
```

The log target prioritizes relative accuracy. The logit target guarantees hard global floor/cap bounds for cliquets.

6. PCHIP, Akima, and piecewise Bezier curves

6.1 Cubic Hermite segment

```
t=(x-x_i)/h, h=x_(i+1)-x_i

H(t)=h00(t)y_i + h10(t)h d_i
     + h01(t)y_(i+1) + h11(t)h d_(i+1)

h00=2t^3-3t^2+1, h10=t^3-2t^2+t
h01=-2t^3+3t^2, h11=t^3-t^2.
```

6.2 PCHIP slope rule

Let secants delta_i=(y_(i+1)-y_i)/h_i. If adjacent secants have opposite sign, PCHIP sets the interior derivative to zero. Otherwise it uses a weighted harmonic mean:

```
d_i = (w1+w2)/(w1/delta_(i-1)+w2/delta_i)
w1=2h_i+h_(i-1), w2=h_i+2h_(i-1).
```

This suppresses overshoot and preserves monotonicity. It is especially useful for American continuation curves near an exercise boundary.

6.2.1 Endpoint slopes and limiting

Interior slopes use information on both sides. At the left endpoint only the first two secants are available, so PCHIP first forms a one-sided estimate:

```
d_0_raw = [(2h_0+h_1) delta_0 - h_0 delta_1] / (h_0+h_1).
```

Then it applies two safety rules. If d_0_raw has the wrong sign relative to delta_0, set d_0=0. If its magnitude exceeds 3|delta_0|, set d_0=3 delta_0. The right endpoint uses the reflected formula.

6.2.2 Why the harmonic mean is shape preserving

Suppose two neighboring secants are positive. Their weighted harmonic mean is also positive and cannot exceed the larger secant by an arbitrary amount. If the secants disagree in sign, the data have a local turning point; setting the derivative to zero prevents the cubic from inventing an extra turn. The formal monotonicity conditions and proof are given in Appendix F.

6.2.3 PCHIP implementation recipe

```
Build once:
1. Sort pairs (x_i,y_i) and remove duplicate x_i.
2. Compute h_i=x_(i+1)-x_i and delta_i=(y_(i+1)-y_i)/h_i.
3. Compute interior d_i with the sign test and harmonic mean.
4. Compute and limit the two endpoint slopes.

Evaluate at x:
5. Find i such that x_i <= x <= x_(i+1).
6. Set t=(x-x_i)/h_i.
7. Evaluate the four cubic Hermite basis functions.
8. Return H_i(t), then invert any log or logit target transform.
9. Override exact payoff, floor, cap, or exercise regions.
```

Building the curve costs O(n). With binary search, each query costs O(log n); on an ordered query grid the interval index can be advanced in O(1) amortized time. PCHIP interpolates every label, so it must be paired with low-noise MC labels; it is a shape-preserving interpolator, not a denoising method.

6.3 Akima slope rule

```
d_i = [w_left delta_(i-1) + w_right delta_i]
      / (w_left+w_right)

w_left  = |delta_(i+1)-delta_i|
w_right = |delta_(i-1)-delta_(i-2)|.
```

Akima weights the less rapidly changing side more heavily. It is local and less aggressively monotone than PCHIP. This matched the reduced Asian curve very well.

6.4 Bezier/Hermite equivalence

```
For the same endpoint slopes:
P0=y_i
P1=y_i + h d_i/3
P2=y_(i+1) - h d_(i+1)/3
P3=y_(i+1).

B(t)=(1-t)^3 P0 + 3(1-t)^2 t P1
     +3(1-t)t^2 P2 + t^3 P3.
```

Expanding B(t) gives the Hermite polynomial exactly. Therefore a piecewise Bezier curve built from PCHIP slopes is PCHIP in another representation, not a distinct smoother.

6.5 Expanded 99-case smoother study

Seventeen distinct one-dimensional estimators were tested over European, American, Asian, and barrier parameter combinations. Every fitted surface used a 10M-scenario training budget; MC benchmarks used 500,000 paths per state. Product families received equal aggregate weight.

Method	Balanced p99	Max for V>=0.05	Local overshoot
Natural cubic interpolation	1.799%	1.163%	2.01%
PCHIP	1.834%	1.220%	0.00%
Akima	1.958%	1.286%	0.17%
MAKIMA	1.968%	1.317%	0.15%
Cubic smoothing spline	2.098%	1.440%	2.49%
Linear interpolation	2.683%	2.058%	0.00%

Natural cubic interpolation led average accuracy by only 0.035 percentage points. PCHIP won 53 of 99 head-to-head p99 cases and introduced no local overshoot, so PCHIP remains the generic one-feature PFE default.

7. European option proxy

7.1 State and benchmark

For constant-parameter GBM, the state at a fixed time is spot alone. The benchmark is the Black-Scholes formula. For a call:

```
d1=[log(S/K)+(r-q+sigma^2/2)tau]/(sigma sqrt(tau))
d2=d1-sigma sqrt(tau)
C=S exp(-q tau) Phi(d1)-K exp(-r tau) Phi(d2).
```

7.2 Why d1 is the fitting coordinate

d1 combines spot, strike, volatility, carry, and remaining maturity into the standardized coordinate controlling exercise probability and delta. Sampling uniformly in d1 is equivalent to deliberate wing coverage in delta space.

7.3 Training label

For each d1 state, shifted antithetic terminal normals produce an unbiased MC label. The default target is log(value), originally fitted by degree-7 Chebyshev ridge.

7.4 One-dimensional method comparison

Method	Worst max error	Average p99	Average MAE
Global Bernstein/Bezier	0.536%	0.208%	0.003546
Chebyshev	0.854%	0.326%	0.004423
Akima	2.540%	1.127%	0.007015
PCHIP	2.554%	1.149%	0.007098
Bezier with PCHIP slopes	2.554%	1.149%	0.007098

Conclusion: the European curve is sufficiently smooth that global regularized bases average MC noise better than exact local interpolation.

For implementation consistency, the standalone universal default is now log-PCHIP in d1. Its independent-seed worst error is 2.637%, versus the product-specific Bernstein result of 0.536%.

8. Arithmetic Asian option proxy

8.1 Raw Markov state

At fixing index j, store current spot S and running sum A of fixings strictly before today. Let N be total fixings and m=N-j-1 future fixings.

```
Payoff = [ (A + S + Sum_(k=1)^m S_(j+k))/N - K ]_+.
```

8.2 Adjusted-strike derivation

```
Payoff
= (m/N) [ (1/m) Sum_(k=1)^m S_(j+k) - K_adj ]_+

K_adj = (N K - A - S)/m.
```

This is an algebraic identity. It rotates the diagonal payoff boundary in (S,A) into a strike-like scalar.

8.3 GBM homogeneity reduction

Future GBM spots satisfy S_(j+k)=S G_k, where the joint growth vector G does not depend on S. Therefore:

```
V(S,A,j)
= (m/N) E[ (S average(G)-K_adj)_+ ]
= S (m/N) E[ (average(G)-K_adj/S)_+ ].
```

At a fixed date the normalized value V/S is a one-dimensional function of K_adj/S, represented by an adjusted d1-like coordinate. This is why a generic 2D fit is unnecessary for this GBM product.

8.4 Exact linear wing

If K_adj<=0 for a call, every future arithmetic average is positive relative to the adjusted strike, so the positive-part operator can be removed. The value is then the discounted expected average minus strike and is computed analytically.

9. Asian benchmark and fitting results

9.1 Geometric Asian control

The log of the discrete future geometric average is a weighted sum of normal increments, hence normal. If weights are w_l=(m-l+1)/m:

```
log(G_geo) = log(S)
 + (r-q-sigma^2/2) dt (m+1)/2
 + sigma sqrt(dt) Sum_l w_l Z_l.

Var(log(G_geo)) = sigma^2 dt Sum_l w_l^2.
```

Thus the geometric Asian call has a Black-Scholes-like exact value and is a highly correlated control for the arithmetic payoff.

9.2 Hybrid target

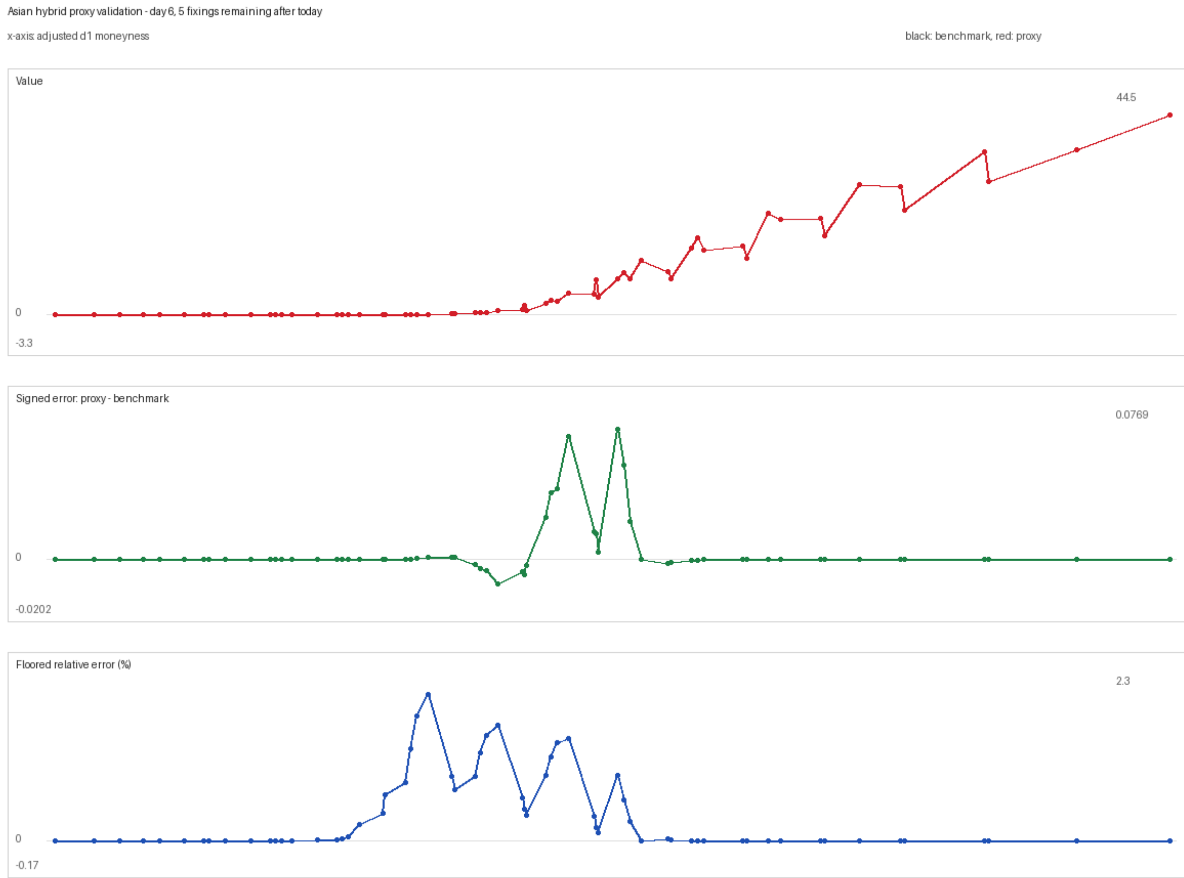
Fit log(V/S) in the OTM and moderate region. In the deep ITM region fit log((V-linear_baseline)/S), then add the exact baseline. This prevents a large intrinsic component from hiding time-value error.

9.3 1D fitter comparison

Method	Worst max error	Average p99	Average MAE
Akima	0.186%	0.057%	0.000178
Chebyshev	3.286%	1.721%	0.013523
PCHIP	3.453%	0.936%	0.000273
Global Bernstein/Bezier	6.854%	3.303%	0.017907
Bezier with PCHIP slopes	3.453%	0.936%	0.000273

Akima is the preferred 1D fitter for this reduced Asian experiment. PCHIP is usable, but its monotonicity limiter can flatten local derivatives when MC labels contain small slope reversals. A second independent-seed run confirmed Akima at 0.144% worst error, versus 2.577% for Chebyshev and 1.643% for PCHIP.

The standalone universal default is nevertheless PCHIP. With its independent default seed it achieved 3.331% worst error while sharing the same fitting engine as European and American options.



Asian adjusted-moneyness proxy diagnostic from the broader experiment.

9A. Ten-asset basket Asian extension

9A.1 Contract and Markov state

The basket Asian experiment prices a monthly arithmetic Asian call on an equal-weight basket of ten correlated GBM assets. Some pairwise correlations are positive and some are negative. At fixing index j the state is the ten-dimensional spot vector S=(S_1,...,S_10) plus the running sum B of previous basket fixings.

```
b(S) = (S_1+...+S_10)/10

Payoff = [ (B + b(S) + Sum_(k=1)^m b(S_(j+k)))/N - K ]_+.
```

Unlike the single-asset GBM Asian, this cannot be reduced exactly to one adjusted-moneyness coordinate because the ten assets have different volatilities, dividends, and correlations. The feature design therefore uses moment information plus PCA directions.

9A.2 Moment baseline

For each state the code computes the first two moments of the final arithmetic basket average exactly under correlated GBM. The identity used is

```
E[S_i(t_a) S_k(t_b)]
= S_i S_k exp(mu_i t_a + mu_k t_b
              + rho_ik sigma_i sigma_k min(t_a,t_b)).
```

Those moments define a moment-matched lognormal approximation. By itself it is smooth and fast, but its day-0 worst max error was 28.450% on the 524,288-path benchmark, so it is used as a baseline rather than the final proxy.

9A.3 PCA sparse-Chebyshev correction

The correction features are log expected-average moneyness, effective average volatility, current basket moneyness, running-average moneyness, cross-sectional dispersion, four principal-component scores of log spots, and log(1+baseline value). A sparse Chebyshev ridge model is fitted to a bounded log-factor correction around the moment baseline.

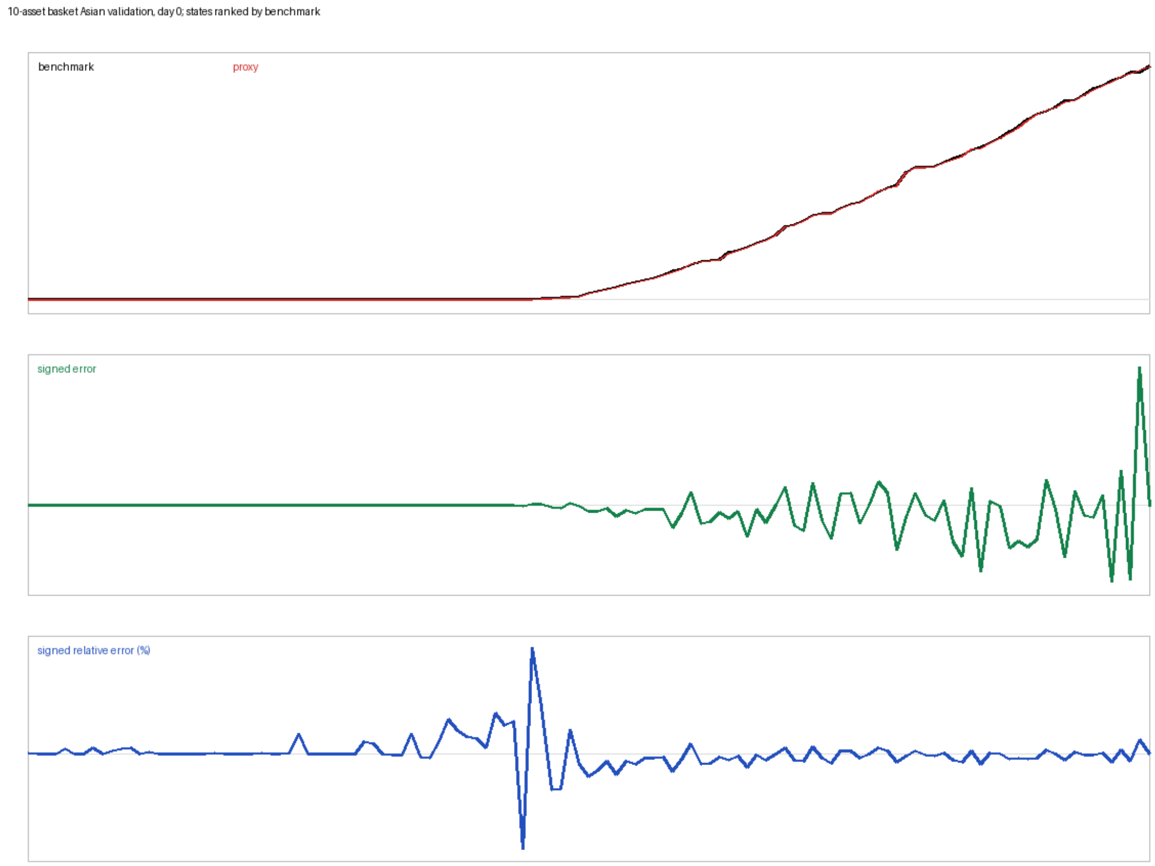
The final default adds a one-dimensional PCHIP calibration of the training residual as a function of expected-average moneyness. This is not an ad hoc hand correction: it is a second-stage smoother trained only on Monte Carlo labels, and it addresses the common situation where a global high-dimensional basis has small but systematic tail bias.

9A.4 Basket Asian result

Method	Worst max error	Average p99	Average MAE
Moment lognormal baseline	28.450%	5.947%	0.023189
Relative residual sparse Chebyshev	13.200%	2.496%	0.018653
Log-factor sparse Chebyshev	8.995%	1.996%	0.012734
PCHIP-calibrated log factor	5.873%	1.820%	0.009023
Raw residual sparse Chebyshev	19.085%	3.339%	0.002380
Fixed residual/log-factor blend	11.452%	2.295%	0.006251

The default therefore meets the requested under-8% worst-error target on the tested state design. Training uses 513 states per date and 65,536 Sobol or Sobol/LR paths per state, or about 33.6 million state-scenarios per date after the power-of-two Sobol rounding. Each validation state uses 524,288 benchmark paths.

Importance-sampling note: the current script uses a two-component Gaussian likelihood-ratio mixture for the long-dated day-0 slice, shifting basket growth toward the exercise boundary and weighting each path by p/q. It uses plain Sobol for later dates because the same shift increased weighted variance there.



Ten-asset basket Asian day-0 diagnostic using the PCHIP-calibrated sparse proxy.

10. American put: optimal stopping

10.1 Snell envelope and Bellman recursion

For exercise dates t_k, intrinsic payoff g(S)=(K-S)_+. The value is the smallest supermartingale dominating g, equivalently the Snell envelope:

```
V_M(S)=g(S)
C_k(S)=exp(-r dt) E[ V_(k+1)(S_next) | S_k=S ]
V_k(S)=max(g(S), C_k(S)).
```

The max creates a moving derivative kink at the exercise boundary. A global polynomial fitted recursively can turn a small local continuation error into a systematic bias at every earlier date.

10.2 MC dynamic programming

At each of 100 exercise dates, the implementation uses 121 log-spaced state nodes, antithetic one-step GBM transitions, and about 10 million transitions over the complete backward pass. It fits C_k, then imposes max(g,C_k) exactly.

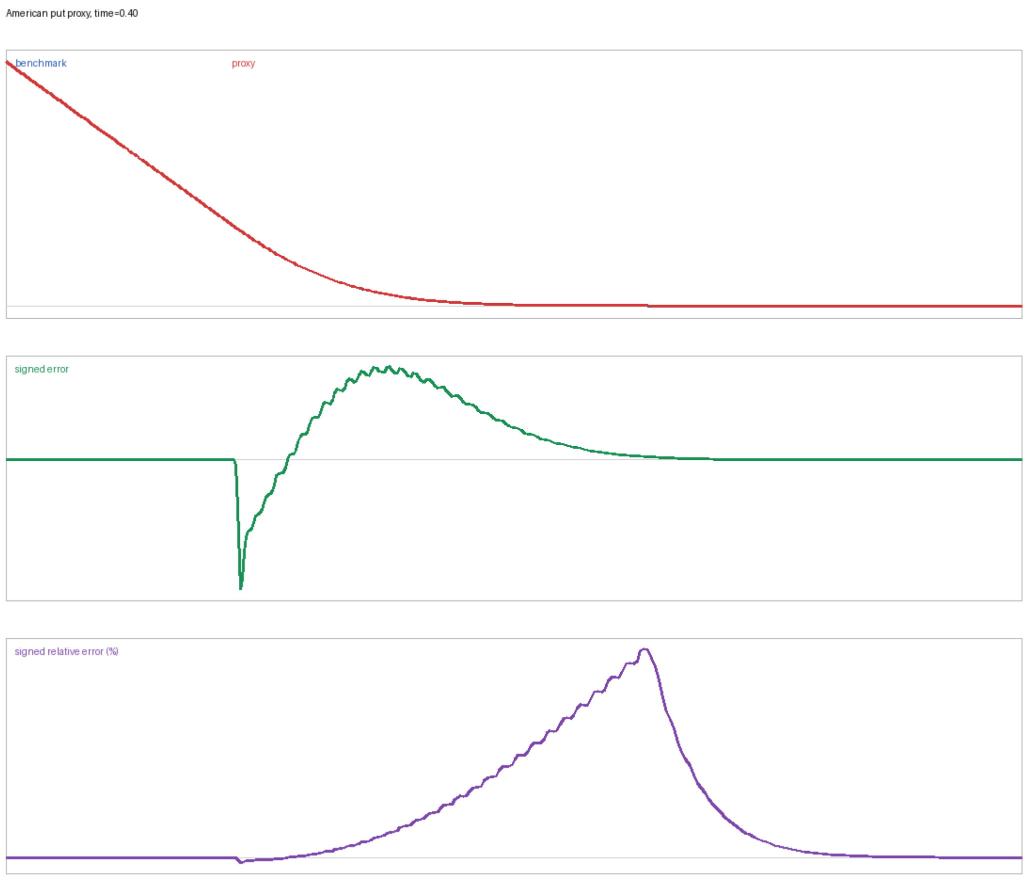
10.3 Why PCHIP wins here

PCHIP is local, shape preserving, and does not ring around the moving boundary. The broad experiment achieved 2.249% worst max error; an independent seed achieved 5.289%. Global log-Chebyshev fits were unstable because exponentiated errors recursively compounded.

Akima was also tested through the complete 100-date recursion. It achieved 3.225% worst max error, 1.675% average p99, and 0.002033 average MAE, versus 2.249%, 0.947%, and 0.001479 for PCHIP. Akima is viable, but PCHIP's monotonicity-preserving slopes are safer near the exercise boundary.

10.4 Independent benchmark

A projected implicit finite-difference solver uses 4,000 time steps and 2,000 spot steps. Projected SOR enforces V>=g. At S=100, t=0, the finite-difference value is about 6.6597 versus 6.6605 from a separate 4,000-step CRR tree.



American put value, signed error, and signed relative error at time 0.4.

10A. Single and double barrier options

10A.1 State and variants

The experiment prices zero-rebate down-and-out, up-and-out, and double-knock-out calls with either monthly discrete or continuous monitoring. Conditional on no historical hit, spot is the only fitted feature. A live/hit indicator must be added once monitoring history exists.

Zero-rebate knock-ins require no second MC training run. Under the same monitoring convention, in/out parity gives V_in = V_vanilla - V_out.

10A.2 Exact Brownian-bridge segment correction

For log endpoints x and y above lower log barrier b over a GBM step with variance q=sigma^2 dt, reflection gives the conditional survival probability

```
P(min bridge > b | x,y)
  = 1 - exp[-2 (x-b)(y-b)/q],  x>b, y>b.
```

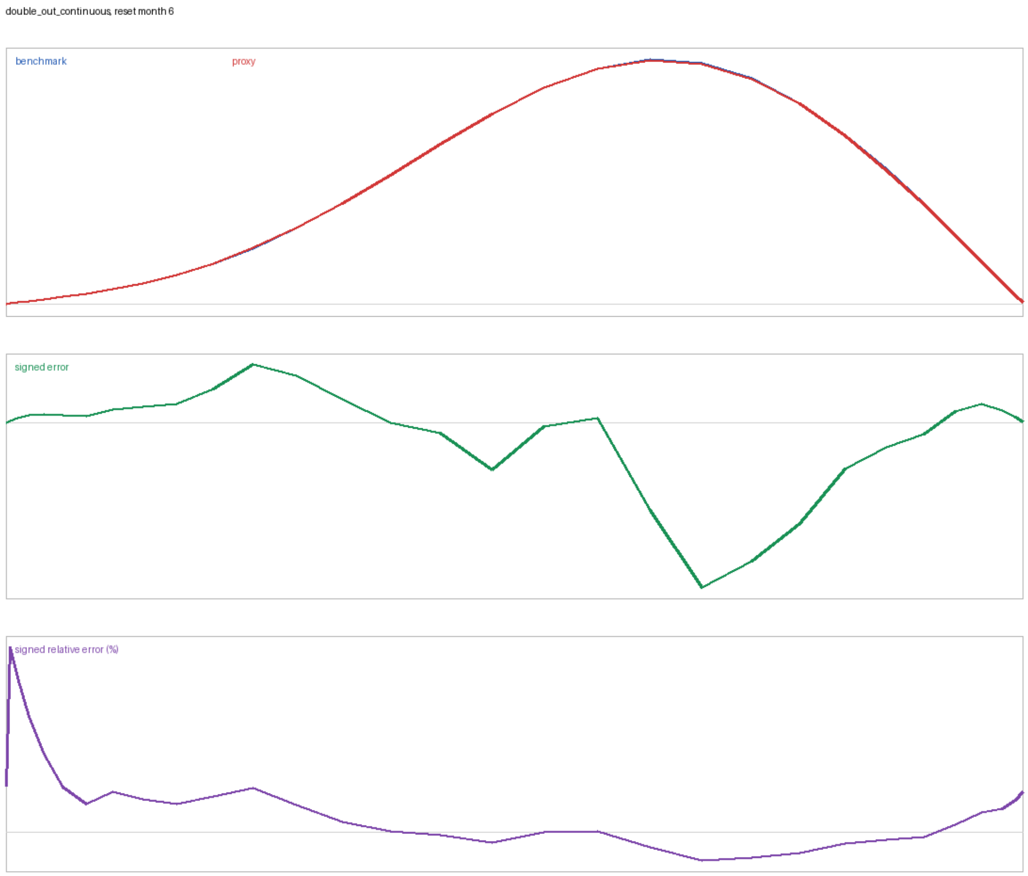
The upper-barrier expression replaces the distances by b-x and b-y. Multiplying segment survival probabilities and weighting the payoff integrates out crossing indicators, reducing variance. For a double barrier, the absorbing interval transition density is evaluated by a method-of-images series and divided by the free transition density.

10A.3 Sampling and results

Log-Chebyshev spot nodes cover the alive domain. Common random numbers make label noise smooth across spot, antithetics reduce path noise, and shifted interior validation nodes prevent training-grid leakage.

Variant	PCHIP raw worst	Worst for V>=0.05
Down-out discrete	2.253%	2.087%
Down-out continuous	2.531%	1.762%
Up-out discrete	9.096%	4.853%
Up-out continuous	8.808%	5.074%
Double-out discrete	8.387%	4.999%
Double-out continuous	8.148%	4.916%

The 8-9% raw maxima are one- or two-cent option values. Meaningful-value errors remain around 2% for down barriers and about 5% for up/double barriers. PCHIP remains the default because the broader 99-case study found near-best accuracy with zero local overshoot.



Continuous double-knock-out value, signed error, and signed relative error.

11. Single-name GBM cliquet

11.1 Payoff

```
c_i = clip(S_i/S_(i-1)-1, local_floor, local_cap)
H = notional * clip(Sum_i c_i, global_floor, global_cap).
```

11.2 State reduction theorem

Under GBM, each future reset return S_i/S_(i-1)-1 depends only on the new normal increment and fixed model parameters, not on S_(i-1). Future coupons are therefore independent of current spot conditional on reset time.

```
State at reset = accrued = Sum of realized clipped coupons.
V = V(accrued, remaining_periods).
```

This is an exact scale-invariance reduction, not an empirical approximation.

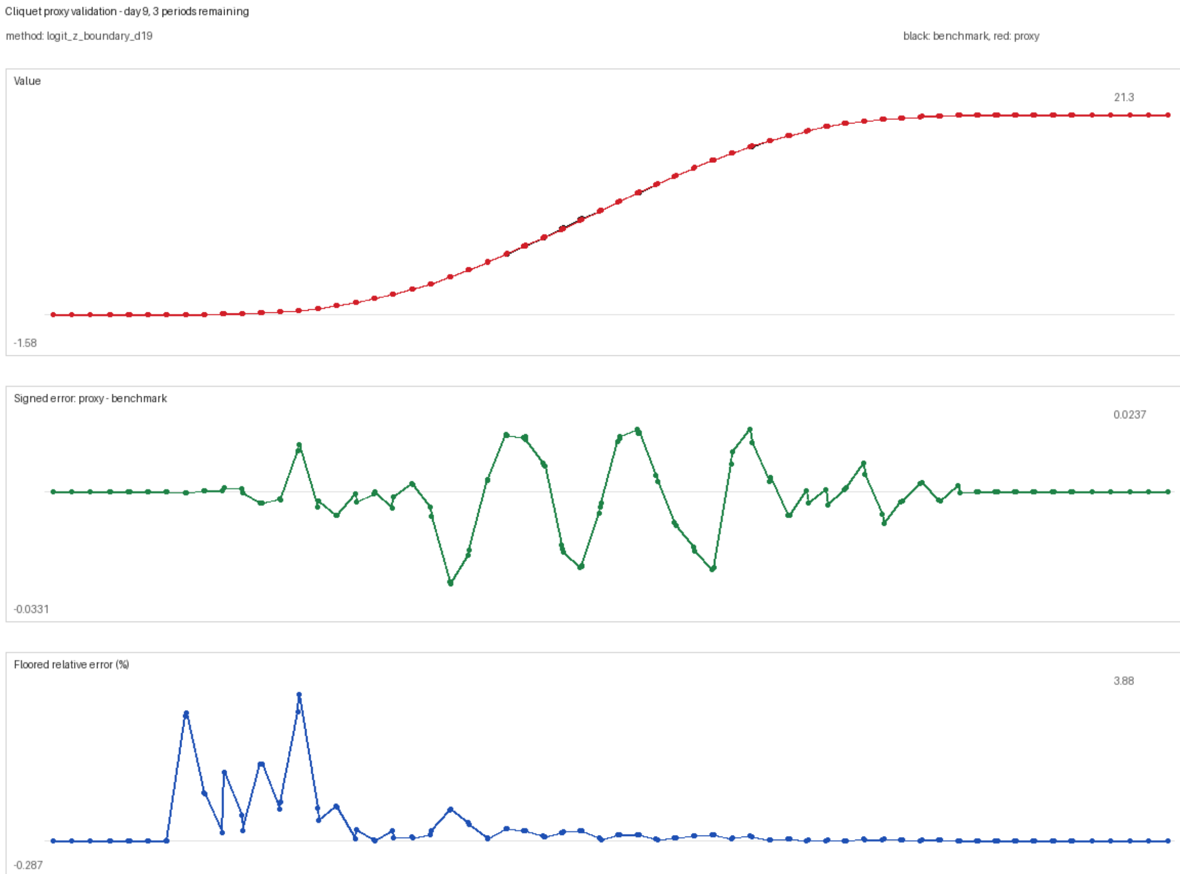
11.3 Expected-total coordinate

```
z = [accrued + m E(c) - midpoint(global bounds)]
    / sqrt(m Var(c)).
```

A boundary-enriched grid samples around the estimated floor and cap transitions. Known flat tails are exact: if even all remaining local caps cannot leave the global floor, or all local floors force the cap, regression is bypassed.

11.4 Result

Bounded-logit Chebyshev degree 19 achieved 3.593% worst max error, 0.928% average p99, and 0.003194 average MAE.



GBM cliquet proxy diagnostic with three coupons remaining.

12. Single-name SLV cliquet

12.1 Expanded state

SLV destroys GBM scale invariance because leverage L(S) depends on spot and future variance depends on current v. The reset state becomes (accrued, S, v).

12.2 Structured features

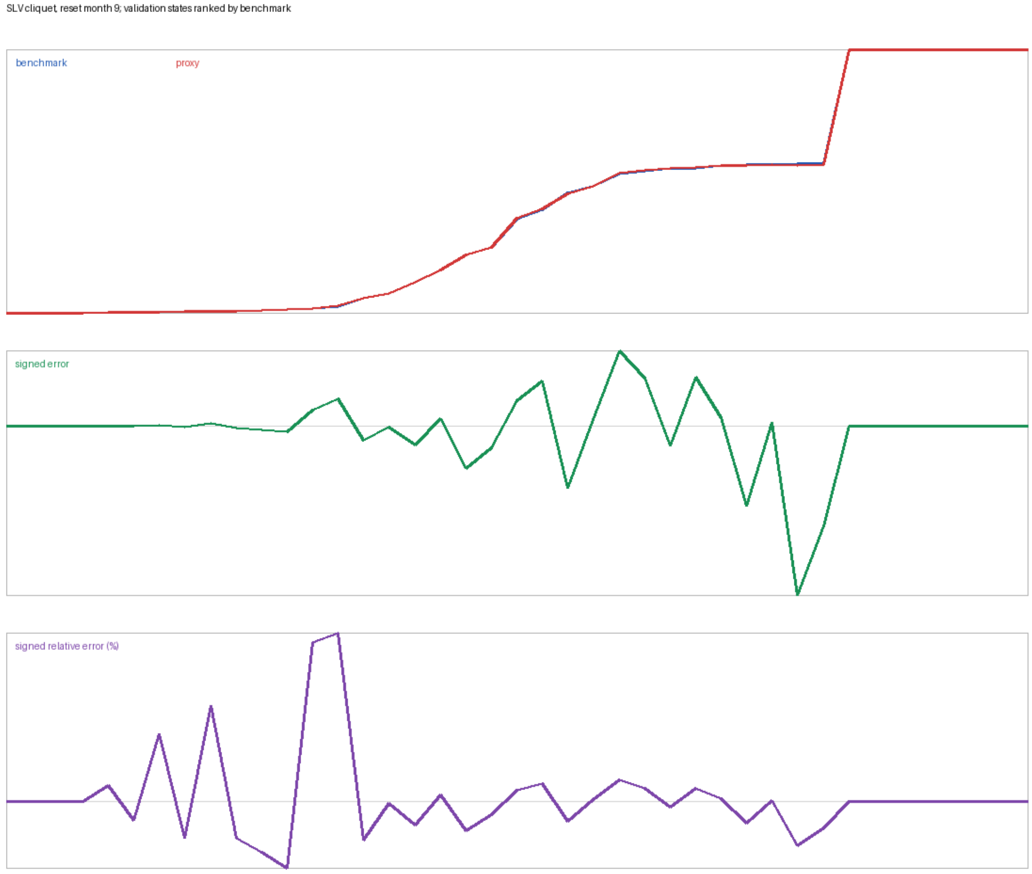
A frozen-coefficient one-period coupon distribution supplies conditional mean and variance. These form an expected-total coordinate z, supplemented by log(S/S0) and log(v/theta). The target remains bounded logit.

12.3 Maturity-adaptive fit

Remaining coupons	Fitter	Reason
9-12	Local quadratic	Broad smooth long-horizon dependence
3-6	Anisotropic Chebyshev	Sharper payoff transitions
0	Exact payoff	No continuation uncertainty

Lower-tail mixture importance sampling was necessary for seed stability. The common-random-number rerun achieved 5.067% worst error. A separate two-design study rebuilt 10M-path labels and used 500,000 benchmark paths per state on both designs; the fixed adaptive hybrid achieved 4.795% worst error.

Anisotropic Chebyshev degrees 13-23, sparse Hermite, local regression, and Nystrom Matern regression were tested in that study. None beat the existing time-based local/spectral rule. Common random numbers are retained because they smooth label noise across state without changing conditional means.



Single-name SLV cliquet validation states ranked by benchmark value.

13. Three-underlying SLV basket cliquet

13.1 Three coupon definitions

Variant	Monthly coupon
Basket return	clip(mean(R1,R2,R3), local floor, local cap)
Average clipped	mean(clip(R1),clip(R2),clip(R3))
Worst of	clip(min(R1,R2,R3), local floor, local cap)

13.2 Seven-dimensional Markov state

```
X = (accrued, S1,S2,S3, v1,v2,v3).
```

Each asset has its own variance factor and leverage function. Correlated market drivers create cross-asset dependence; each spot shock also contains its own negatively correlated variance shock.

```
Z_i^S = rho_i Z_i^v + sqrt(1-rho_i^2) Z_i^M,
Corr(Z^M)=R_market.
```

This construction is automatically positive semidefinite because it is a linear transformation of independent variance normals and a valid correlated market-normal vector.

13.3 Sampling

2,001 low-discrepancy boundary-enriched states cover accrued return, three log spots, and three log variances. Approximately 10 million scenarios are spread over each reset-date fit. Each of 31 independent validation states uses 500,000 paths.

13.4 Features and bounded target

Conditional coupon moments define separate lower and upper standardized cushions. Keeping both is important: one midpoint z cannot distinguish states with equal standardized midpoint but different dispersion and distances to the two global bounds.

14. Basket high-dimensional model search

The search compared local full-state and summary regressions, sparse anisotropic Chebyshev terms, Gaussian RBF kernels, and a two-layer tanh neural ensemble. With averaged state labels, neither RBF nor the small neural ensemble was uniformly competitive. This does not reject neural networks in general; it shows that architecture and state coverage matter.

14.1 Development ensemble and validation failure

```
V_proxy = w(m) V_local + [1-w(m)] V_spectral
w(m) = 0.16 + 0.02 m,
m = remaining coupons.
```

For symmetric basket-return and average-clipped coupons, V_local used summary features. Worst-of used the full state. The blend performed well on the first development design but did not generalize, so it was retired.

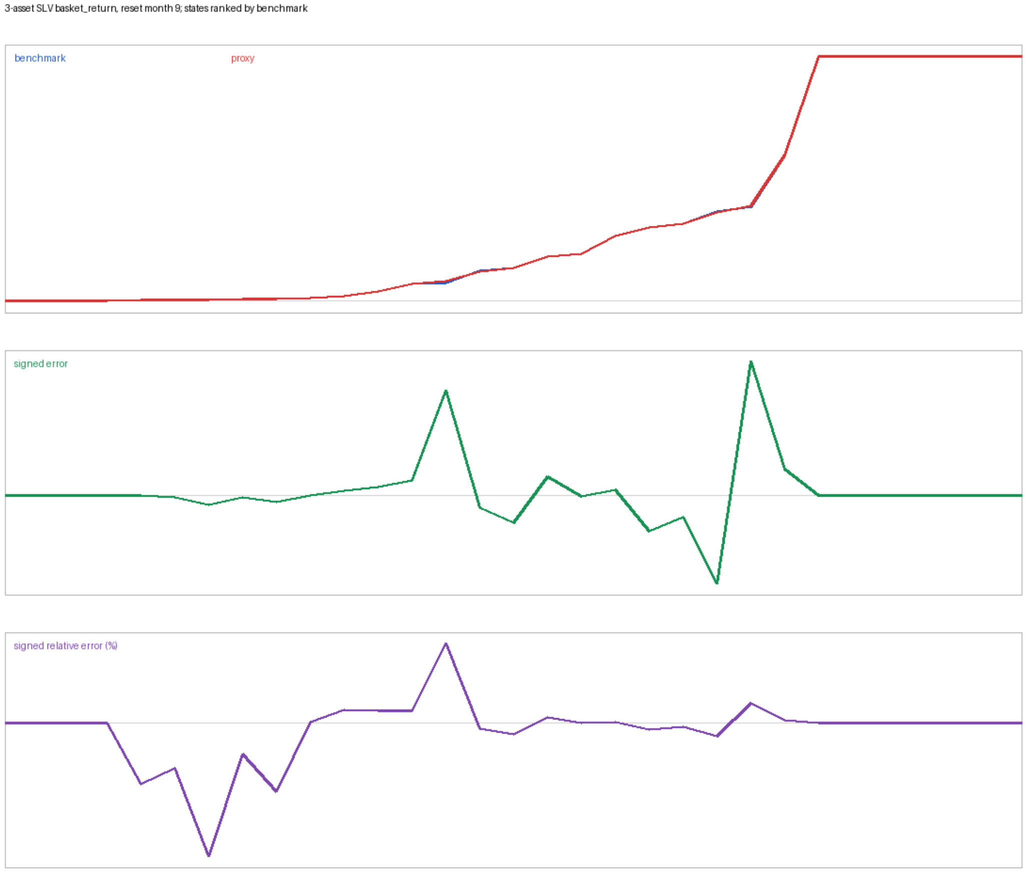
Variant	Development	Validation 2	Validation 3	Worst
Basket return	10.258%	7.108%	18.500%	18.500%
Average clipped	5.624%	11.566%	5.745%	11.566%
Worst of	9.700%	9.862%	6.774%	9.862%

These are fixed sparse-spectral results using 2,001 states. A 5,001-state design, local models, RBF kernels, small tanh networks, random-feature networks, and bagging were also tested. None guaranteed 5-8% on the third untouched design. The next credible step is adaptive state enrichment around failed neighborhoods, followed by another untouched validation design.

14.2 Literature-inspired residual search

A clipped-normal moment baseline was used as a low-fidelity model, followed by direct or residual sparse Hermite, local, Nystrom Matern, and fixed ensemble corrections. Coupon skewness and floor/cap masses were added as features. Inverse-variance weighted sparse Chebyshev fits were also tested on both 2,001 and 5,001 states.

None improved worst-case relative error on the development and independent designs. Some dense weighted fits reduced MAE but leaked larger errors into tail states. The moment baseline did not simplify correlated clipped and worst-of SLV coupons enough for residual learning to help.



Development diagnostic for the fixed sparse basket-return proxy with three coupons remaining.

15. Generic workflow for a new instrument

Step 1: define the contract exactly

- Observation schedule, exercise rights, local/global bounds, and settlement.

- Clarify basket aggregation order: clip-then-average is not average-then-clip.

Step 2: identify a sufficient Markov state

- Start from the simulator state and add path statistics required by payoff.

- Prove any dimension reduction from scale invariance or payoff algebra.

Step 3: nondimensionalize

- Use moneyness, normalized variance, expected-total cushions, and time.

- Align transition boundaries across maturities before fitting.

Step 4: exploit exact regions

- Terminal payoff, linear ITM wings, hard global floor/cap tails, intrinsic value.

Step 5: design state sampling

- Combine broad low-discrepancy coverage with boundary-focused clusters.

- Do not spend most paths in flat regions that are already known analytically.

Step 6: generate MC labels

- Use antithetic paths by default.

- Add controls when exact expectations are available.

- Attempt likelihood-ratio shifts for rare payoff wings whenever p/q is known and numerically stable.

- Keep the shift only on slices where the weighted estimator lowers variance or tail error.

- Keep state-space enrichment distinct from true importance sampling in the run notes.

Step 7: choose fitter by effective dimension

Effective dimension	Default candidates
Any genuine 1D price proxy	PCHIP operational default
1D smooth, product optimized	Natural cubic, Chebyshev, Bernstein
1D noisy labels	Smoothing spline or P-spline after validation
2-5D smooth	Sparse anisotropic basis, local polynomial, small NN
6D+	Smooth NN or sparse/additive model; ensemble after validation

Step 8: validate independently

- Use new random seeds and a stronger benchmark.

- Report value, signed error, relative error, and benchmark standard error.

- Slice results by maturity, payoff region, and state-space boundary.

- For PFE, run the proxy through outer scenarios and compare exposure quantiles by date.

15A. Instrument-by-instrument proxy recipes

Each row below is a reproducible recipe. The state is what must be known at the valuation date; the label column describes the expensive price calculation; the transform and fitter define the cheap proxy.

Instrument	State / coordinate	Training label	Target and fitter
European	Spot S; transform to d1	Shifted antithetic terminal GBM payoff	log(V+eps), PCHIP
Asian	(S,A); reduce to adjusted moneyness	Path MC + geometric-Asian control	log(V/S) and log(time/S), PCHIP
10-asset basket Asian	10 spots plus running basket sum	Sobol path MC; moment baseline	PCA sparse-Chebyshev + PCHIP residual calibration
American	Spot S at each exercise date	One-step MC conditional continuation	Raw continuation PCHIP; max with intrinsic
Barrier	Spot S plus alive/hit flag	Antithetic MC + bridge survival weight	log(V+eps), PCHIP

Recipe details: European

- Choose a valuation date and compute remaining maturity tau.

- Generate spot nodes uniformly in d1 so both delta wings are covered.

- At each node, importance-shift terminal normals toward the strike.

- Fit log price with PCHIP; use the payoff exactly at maturity.

- Validate against Black-Scholes on a denser shifted grid.

Recipe details: arithmetic Asian

- Store spot S and running sum A before today's fixing.

- Derive K_adj=(N K-A-S)/m and the scalar adjusted-moneyness coordinate.

- Use antithetics, likelihood shifting, and the exact geometric Asian control.

- Fit normalized value and time value; use the exact linear wing.

- Validate with 500,000 independent paths per state.

Recipe details: ten-asset basket Asian

- Store all ten spots and the running sum of previous equal-weight basket fixings.

- Compute exact first and second moments of the final arithmetic basket average.

- Use the moment-matched lognormal value as a baseline, not as the final proxy.

- Add PCA scores of log spots to represent cross-sectional composition.

- Fit bounded sparse-Chebyshev corrections, then calibrate residual bias with PCHIP.

- Use a Gaussian likelihood-ratio mixture for long-dated OTM slices when it lowers weighted variance.

- Fall back to plain Sobol on slices where the proposed shift increases variance.

15B. Exercise, barrier, and cliquet recipes

Recipe details: American put

- Start from the terminal payoff and move backward through exercise dates.
- Simulate one-step transitions at 121 spot nodes.
- Average the discounted next-date proxy to estimate continuation.
- Fit continuation with PCHIP, then impose V=max(intrinsic, continuation).
- Validate against an independent projected finite-difference solution.

Recipe details: barrier call

- Condition on whether the barrier has already been hit.
- For discrete monitoring, check simulated prices only at monitoring dates.
- For continuous monitoring, multiply exact segment bridge-survival weights.
- Fit the alive-state log price in spot; obtain zero-rebate knock-in by parity.

Recipe details: cliquet family

Instrument	State	Variance reduction	Proxy
GBM cliquet	Accrued clipped return	Antithetic + clipped-sum control	Bounded logit Chebyshev d19
Single SLV	Accrued, spot, variance	Antithetic + lower-tail shift	Local / anisotropic Chebyshev
3-asset SLV	Accrued, 3 spots, 3 variances	Antithetic + mixture shift	Sparse bounded Chebyshev

- Compute exact global-floor and global-cap states before fitting.
- Transform the remaining bounded price to logit space.
- Concentrate states around floor/cap transition regions.
- For the 7D basket, validate on multiple independent state-space designs.

The 7D basket is the only family that did not reliably meet the 5-8% maximum-error goal. The documented default is the strongest fixed baseline, not a claim that the dimensionality problem has been solved.

16. Why there is no universal fitter

PCHIP works well for the American continuation curve because the geometry is one-dimensional, monotone, and kinked. It is not directly defined on a general multidimensional cloud. Tensor-product PCHIP would grow exponentially with dimension and requires a structured grid.

Chebyshev and Bernstein bases are excellent global smoothers in one dimension. In seven dimensions, unrestricted tensor bases are combinatorial. Sparse term selection helps, but sharp payoff transitions and interactions can still produce bias.

Neural networks are a genuinely generic high-dimensional candidate because parameter count can grow without a tensor grid. Their success requires enough state diversity, regularization, architecture tuning, and independent seeds. The small basket network tested here did not beat the structured ensemble.

A robust universal *workflow* is more realistic than a universal fitter: state reduction, payoff-aware features, bounded targets, variance-reduced labels, dimension-appropriate smoothers, and independent validation.

Recommended hierarchy

Question	Action
Can the state be proved 1D?	Use and compare global ridge, Akima, PCHIP.
Are hard bounds known?	Use exact tails and bounded-logit target.
Is there a dominant baseline?	Fit residual or time value.
Do local/global models err oppositely?	Use a simple cross-validated blend.
Does dimension exceed sparse-basis capacity?	Increase state coverage and test smooth NN.

17. Error accounting

17.1 Metrics

```
signed_error = proxy - benchmark
absolute_error = |signed_error|
relative_error = absolute_error / max(|benchmark|, 0.01)
noise_ratio = absolute_error / max(benchmark_SE, tiny).
```

A low noise ratio indicates the discrepancy may be statistically unresolved. A large noise ratio indicates model/fitting bias rather than benchmark noise.

17.2 Error decomposition

```
proxy - true continuous-model value
= fitting error
 + training MC error
 + benchmark estimation comparison noise
 + time-discretization bias
 + model/calibration error.
```

The reported studies isolate fitting plus training noise against a benchmark under the same model discretization. They do not claim market calibration accuracy.

17.3 Leakage controls

- Independent training and benchmark random streams.

- Fixed validation states not reused as training states.

- A final standalone run with an independent seed where provided.

- Method-selection results should be confirmed on a second validation design.

17.4 Stability checks

- Repeat labels with different seeds.

- Double Euler steps for SLV and compare.

- Expand spot/variance domains to test extrapolation.

- Check price bounds and monotonicity numerically.

- For Greeks, validate derivative smoothness separately from price error.

18. Reproducibility map

Purpose	Entry point or folder
European default	EuroMain.py
Asian default	AsianMain.py
10-asset basket Asian	BasketAsianMain.py
Barrier research/default	BarrierOptExperiment/BarrierMain.py
GBM cliquet default	CliquetMain.py
American put default	AmericanMain.py
Single-name SLV cliquet	SLVCliquetMain.py
3-asset SLV basket cliquet	BasketCliquetOptExperiment/BasketCliquetMain.py
Expanded 1D comparison	OneDimensionalFitExperiment/ExpandedSplineStudy.py
European research	EuroOptExperiment/
Asian research	AsianOptExperiment/
Cliquet research	CliquetOptExperiment/
American research	AmericanOptExperiment/
SLV research	SLVCliquetOptExperiment/ and BasketCliquetOptExperiment/

Core numerical budgets

Study	Training	Benchmark
European	121 states x 25,000 shifted paths	Black-Scholes closed form
Asian	about 10M scenarios per fitted date	500K paths/state + geometric control
10-asset basket Asian	513 states x 65,536 Sobol/LR paths/state	524,288 Sobol/LR paths/state
GBM cliquet	about 10M scenarios per fitted date	500K paths/state + clipped-sum control
American	10,006,700 one-step transitions	4,000 x 2,000 projected FD grid
Barrier	10M scenarios per fitted date	500K paths/state + bridge survival
Single SLV cliquet	about 10M scenarios per fitted date	500K paths/state
Basket SLV cliquet	about 10M scenarios over 2,001 states/date	500K paths x 31 states/date

Known limitations

- Illustrative SLV leverage functions are not calibrated market surfaces.

- SLV uses two Euler steps per monthly period in the current experiments.

- Basket Asian uses conditional LR mixture IS; it is deliberately off on dates where it raised variance.

- Basket 5-8% control failed on untouched designs; adaptive enrichment remains research.

- Pointwise price accuracy does not replace outer-scenario PFE quantile validation.

- Production extrapolation, Greeks, and calibration loops require separate tests.

- The neural basket test is not a definitive neural-network architecture search.

19. Final conclusions

The strongest improvement repeatedly came from changing the representation before changing the regressor. d1 organizes European wings; adjusted strike reduces the GBM Asian state; expected-total cushions organize cliquet bounds; the Snell recursion isolates American continuation.

The expanded one-dimensional test covered 99 European, American, Asian, and barrier cases with 17 estimators. Natural cubic interpolation led balanced p99 by only 0.035 percentage points, while PCHIP won 53 of 99 head-to-head cases and had zero local overshoot.

When one implementation must cover every genuinely one-dimensional option, PCHIP is therefore the universal default. Product-specific optimized profiles remain available when maximum pointwise accuracy matters. For PFE, the final acceptance metric must be exposure-quantile distortion in actual outer scenarios, not price RMSE alone.

For higher-dimensional SLV baskets, sparse payoff-aware regression was the strongest generic baseline, but three independent state designs showed that the 10M-path budget did not guarantee the 5-8% target. Robust worst errors were approximately 18.5%, 11.6%, and 9.9%.

For the ten-asset basket Asian, the successful generic pattern was different: use an analytically moment-matched low-fidelity baseline, summarize the remaining composition with PCA, fit a sparse Chebyshev log-factor correction, and then use PCHIP as a one-dimensional residual calibrator. Conditional LR mixture importance sampling improved the long-dated OTM slice. That fixed default achieved 5.873% worst max error against 524,288-path benchmarks.

Moment-residual Hermite, local, Nystrom Matern, weighted spectral, and fixed ensemble alternatives did not improve the 7D basket tail. This supports adaptive state enrichment or a larger path-level neural program rather than further tuning on the same sparse labels.

The reusable methodology is therefore: derive the state, reduce dimension when mathematically valid, enforce exact structure, create unbiased low-noise labels, fit with a dimension-appropriate smoother, and validate independently in the wings and transition regions.

The remaining pages are theorem statements, proofs, and references. They are not required to run the code, but they explain why each step is mathematically valid.

Appendix A. Probability and Monte Carlo theorems

The theorem statements below are restated in the notation of this project; they are not verbatim quotations. Each statement lists the assumptions used by the implementation.

A.1 Sample mean and standard error

Theorem (Law of Large Numbers and Central Limit Theorem). Let Y_1,Y_2,... be independent, identically distributed path values with finite mean mu and finite nonzero variance s^2. Then the sample mean converges to mu in probability. Moreover, sqrt(N)(Y_bar-mu)/s converges in distribution to a standard normal random variable.

Proof. The weak law follows from Var(Y_bar)=s^2/N and Chebyshev's inequality: P(|Y_bar-mu|>a) <= s^2/(N a^2), which tends to zero. The central-limit statement follows by expanding the characteristic function of the centered, standardized variable near zero and raising it to the Nth power. Replacing s by the sample standard deviation is valid because that estimator is consistent.

A.2 Conditioning and variance reduction

Theorem (Tower Property and Rao-Blackwell Variance Identity). For square-integrable Y and information G, E[E[Y|G]]=E[Y] and Var(Y)=E[Var(Y|G)]+Var(E[Y|G]). Therefore Var(E[Y|G])<=Var(Y).

Proof. The defining property of conditional expectation gives the first identity by choosing the whole sample space as the event. Write Y-E[Y]=(Y-E[Y|G])+(E[Y|G]-E[Y]). The cross term has expectation zero after conditioning on G, so expanding the square gives the variance identity. Brownian-bridge survival weighting uses exactly this idea.

A.3 Control variates

Theorem (Optimal Linear Control). Let Y be the discounted target payoff and C a control with known mean m_C and positive variance. The estimator Y_beta=Y-beta(C-m_C) is unbiased for every beta. Its variance is minimized by beta*=Cov(Y,C)/Var(C).

Proof. Unbiasedness follows because E[C-m_C]=0. Expanding the variance gives Var(Y_beta)=Var(Y)+beta^2 Var(C)-2 beta Cov(Y,C), a convex quadratic. Differentiating and setting the derivative to zero gives beta*.

A.4 Gaussian likelihood-ratio shift

Theorem (Normal Mean Shift). If Z is sampled from N(theta,1), then for any integrable payoff f, E_N(0,1)[f(Z)]=E_N(theta,1)[f(Z) exp(-theta Z+theta^2/2)].

Proof. Divide the standard-normal density phi(z) by the shifted density phi(z-theta). Completing the square gives their ratio exp(-theta z+theta^2/2). Multiplying the shifted integral by this ratio recovers the original integral exactly. The multistep formula multiplies these ratios, which adds their log likelihoods.

A.5 Antithetic unbiasedness

Theorem (Antithetic Pairing). If Z and -Z have the same distribution, then [f(Z)+f(-Z)]/2 has the same expectation as f(Z). Its variance is no greater than plain two-path averaging when Cov(f(Z),f(-Z))<=0.

Proof. Symmetry gives E[f(-Z)]=E[f(Z)]. The paired variance is one half of Var(f(Z)) plus one half of the covariance term, so a nonpositive covariance cannot increase variance.

Appendix B. Risk-neutral pricing and GBM

B.1 Risk-neutral pricing theorem

Theorem (Risk-Neutral Valuation). In an arbitrage-free complete market with money-market numeraire B_t, there is a unique probability measure Q under which every traded discounted price S_t/B_t is a martingale. A replicable payoff H at T has value V_t=B_t E_Q[H/B_T | F_t].

Proof. A self-financing replicating portfolio has discounted value equal to a Q-martingale, so its current value is the conditional expectation of its discounted terminal value. At T that terminal value equals H. If the option had a different price, buying the cheaper claim and selling the dearer replicating strategy would create an arbitrage. Existence and uniqueness of Q are the finite-market content of the fundamental theorem of asset pricing.

B.2 Ito's lemma

Theorem (Ito's Lemma, One Dimension). If X_t satisfies dX_t=a(t,X_t)dt+b(t,X_t)dW_t and f has one continuous time derivative and two continuous space derivatives, then df=(f_t+a f_x+0.5 b^2 f_xx)dt+b f_x dW_t.

Proof. Use a second-order Taylor expansion over a short interval. Brownian increments have size sqrt(dt), so (dW)^2 contributes at order dt, while dt*dW and dt^2 are smaller. The quadratic variation identity (dW)^2=dt produces the extra 0.5 b^2 f_xx term. A rigorous proof takes limits in probability over partitions; that measure-theory step is beyond the assumed prerequisites.

B.3 Exact GBM solution

Theorem (Geometric Brownian Motion). If dS/S=(r-q)dt+sigma dW with constant coefficients, then over tau, S_T=S_t exp[(r-q-sigma^2/2)tau+sigma sqrt(tau)Z], Z~N(0,1).

Proof. Apply Ito's lemma to f(S)=log S. Since f'=1/S and f''=-1/S^2, d log S=(r-q-sigma^2/2)dt+sigma dW. Integrate from t to T and use W_T-W_t=sqrt(tau)Z, then exponentiate.

B.4 Black-Scholes call formula

Theorem (Black-Scholes-Merton Call). Under the GBM assumptions above, a European call has value C=S exp(-q tau)Phi(d1)-K exp(-r tau)Phi(d2), where d1=[log(S/K)+(r-q+sigma^2/2)tau]/(sigma sqrt(tau)) and d2=d1-sigma sqrt(tau).

Proof. Risk-neutral pricing gives exp(-r tau)E[(S_T-K)+]. Split the expectation into E[S_T 1_(S_T>K)]-K P(S_T>K). The lognormal threshold is Z>-d2, giving P=Phi(d2). Completing the square inside the first normal integral shifts the threshold by sigma sqrt(tau), giving E[S_T 1]=S exp((r-q)tau)Phi(d1). Discounting produces the formula.

B.5 Correlated GBM second moment

Theorem (Two-Asset GBM Moment). Let S_i(t)=S_i(0) exp(mu_i t+sigma_i W_i(t)) with Corr(dW_i,dW_k)=rho_ik. Then E[S_i(t_a)S_k(t_b)]=S_i(0)S_k(0) exp(mu_i t_a+mu_k t_b+rho_ik sigma_i sigma_k min(t_a,t_b)).

Proof. The product equals S_i(0)S_k(0) times the exponential of a normal variable X=mu_i t_a+mu_k t_b+sigma_i W_i(t_a)+sigma_k W_k(t_b). Its variance is sigma_i^2 t_a+sigma_k^2 t_b+2 rho_ik sigma_i sigma_k min(t_a,t_b). For normal X, E[exp(X)]=exp(E[X]+0.5 Var(X)). In the risk-neutral GBM parameterization the log drift mu_l already includes -0.5 sigma_l^2, so the two single-asset 0.5 variance terms cancel those drift corrections, leaving exactly the cross term shown above.

Appendix C. Path-dependent identities and barriers

C.1 Asian homogeneity reduction

Theorem (GBM Asian State Reduction). At a fixed fixing date with m future fixings, define K_adj=(N K-A-S)/m. Under GBM, the arithmetic Asian call value satisfies V(S,A)=S times a function only of K_adj/S and time.

Proof. Future spots can be written S G_k, where the joint growth factors G_k do not depend on S. The payoff equals (m/N)[S average(G)-K_adj]+. Pulling S>0 outside the positive part gives S(m/N)[average(G)-K_adj/S]+. Taking the discounted expectation leaves a scalar function of K_adj/S.

C.2 Lognormal geometric Asian control

Theorem (Weighted Normal Sum). Any fixed linear combination of jointly normal variables is normal. Therefore the log of a discrete geometric average of GBM fixings is normal.

Proof. Each future log spot is an affine function of Gaussian increments. Their average is another affine combination of the same increments, hence normal by the defining closure property of multivariate normal vectors. Exponentiating makes the geometric average lognormal, so its option value is available in Black-Scholes form.

C.3 In/out parity

Theorem (Zero-Rebate Barrier Parity). For identical strike, maturity, barrier, and monitoring, a zero-rebate knock-in plus the corresponding knock-out equals the vanilla option path by path. Hence V_in+V_out=V_vanilla.

Proof. Every path is in exactly one of two disjoint events: the barrier is hit or it is not. On a hit path only the knock-in pays the vanilla payoff; on a no-hit path only the knock-out pays it. Add the two path payoffs and then take the discounted expectation.

C.4 Brownian-bridge survival

Theorem (Single Lower-Barrier Bridge). Let a Brownian bridge with variance rate sigma^2 start at x>b and end at y>b after dt. Its probability of staying above b is 1-exp[-2(x-b)(y-b)/(sigma^2 dt)].

Proof. For Brownian motion starting at x, the reflection principle maps every path that first hits b and ends at y>b to a path ending at the reflected point 2b-y. The ratio of the reflected Gaussian transition density to the direct density is exp[-2(x-b)(y-b)/(sigma^2 dt)]. Conditioning on the endpoint turns this ratio into the crossing probability; subtract from one.

C.5 Double-barrier image expansion

Theorem (Absorbing Interval Kernel). For Brownian motion killed at lower a and upper c, the transition density inside (a,c) equals an alternating infinite sum of free Gaussian densities at repeatedly reflected image points. Dividing this density by the free transition density gives conditional double-barrier survival.

Proof. Reflect the endpoint across a to enforce zero density at the lower boundary, then repeat reflections every 2(c-a) to enforce the upper boundary as well. At either boundary the direct and reflected terms cancel pairwise. Each term solves the heat equation, and the sum has the correct initial mass; uniqueness of the absorbing heat-equation solution identifies the kernel. The code truncates the rapidly decaying image sum and clips roundoff to [0,1].

Appendix D. American exercise and exact payoff bounds

D.1 Bellman recursion

Theorem (Finite-Horizon Optimal Stopping). For exercise dates $k=0,...,M$ and discounted reward g_k, define $V_M=g_M$ and $V_k=\max(g_k, E[V_{(k+1)}|F_k])$. Then V_k is the largest value obtainable by any stopping rule from date k, and the first date where $g_k>=E[V_{(k+1)}|F_k]$ is optimal.

Proof. Use backward induction. At the last date, stopping is the only action. Assume $V_{(k+1)}$ is optimal from the next date. At k there are only two choices: stop for g_k, or continue and receive conditional expected value $E[V_{(k+1)}|F_k]$. Taking their maximum is therefore optimal. Induction proves the result at all dates. This is the discrete Snell-envelope construction.

D.2 Cliquet floor and cap tails

Theorem (Bound Propagation). If m coupons remain and every coupon lies in $[l,u]$, then final accumulated return lies in $[a+m\,l, a+m\,u]$. If this whole interval is below the global floor or above the global cap, the discounted payoff is known exactly.

Proof. Adding m quantities each between l and u adds between $m\,l$ and $m\,u$. If the largest possible final return is below the floor, clipping always returns the floor. If the smallest possible final return is above the cap, clipping always returns the cap. No simulation is needed in either region.

D.3 Positive semidefinite factor construction

Theorem (Factor-Generated Covariance). If epsilon has identity covariance and $Z=A\,\epsilon$, then $Cov(Z)=A A'$ is positive semidefinite.

Proof. For every vector c, $c'Cov(Z)c=c'A A'c=|A'c|^2>=0$. The basket SLV shock construction is a linear factor model of this form, so it cannot create an invalid negative-variance direction.

Appendix E. Regression and sparse Chebyshev proofs

E.1 Ridge regression solution

Theorem (Ridge Normal Equation). For objective J(beta)=|A beta-y|^2+lambda|Gamma beta|^2, every minimizer satisfies (A'A+lambda Gamma'Gamma)beta=A'y. If that matrix is positive definite, the minimizer is unique.

Proof. Differentiate the quadratic: grad J=2A'(A beta-y)+2lambda Gamma'Gamma beta. Setting it to zero gives the equation. The Hessian is twice the coefficient matrix. Positive definiteness makes J strictly convex, so its stationary point is the unique global minimizer.

E.2 Chebyshev recurrence and boundedness

Theorem (Chebyshev Cosine Identity). For x=cos(theta), T_n(x)=cos(n theta). Consequently |T_n(x)|<=1 on [-1,1] and T_(n+1)=2x T_n-T_(n-1).

Proof. The cosine addition identity gives cos((n+1)theta)=2cos(theta)cos(n theta)-cos((n-1)theta), proving the recurrence. The bound follows because an ordinary cosine lies in [-1,1]. This bounded basis is numerically safer than powers x^n on a wide raw domain.

E.3 Curse of the full tensor

Theorem (Tensor Term Count). If each of D coordinates may use degrees 0 through p independently, the full tensor polynomial basis contains (p+1)^D terms.

Proof. There are p+1 choices for each coordinate degree alpha_j. The multiplication principle gives (p+1) multiplied by itself D times. For D=7 and p=5 this is 6^7=279,936 terms before any regression is solved.

E.4 Sparse anisotropic index sets

Theorem (Downward-Closed Sparse Set). If I={alpha: Sum_j w_j alpha_j<=p} with positive weights, then alpha in I and 0<=beta_j<=alpha_j imply beta in I.

Proof. Because all weights are positive, Sum_j w_j beta_j<=Sum_j w_j alpha_j<=p. Thus lower-order parent terms are retained whenever a higher-order term is retained, which makes the sparse hierarchy interpretable.

Appendix F. PCHIP and cubic interpolation theorems

F.1 Unique cubic Hermite segment

Theorem (Hermite Interpolation). Given two distinct endpoints, two endpoint values, and two endpoint slopes, there is exactly one polynomial of degree at most three satisfying all four conditions.

Proof. The four Hermite basis functions displayed in Section 6 construct one such polynomial. If two cubics satisfied the conditions, their difference would have a double root at each endpoint, hence at least four roots counting multiplicity. A nonzero cubic cannot have four roots, so the difference is zero.

F.2 Local PCHIP monotonicity

Theorem (Fritsch-Carlson / Fritsch-Butland Monotone Cubic). For strictly ordered nodes with monotone data, choosing interior derivatives by the sign test and weighted harmonic mean, together with the stated endpoint limiters, produces a monotone piecewise cubic Hermite interpolant.

Proof. On one interval, subtract y_i and divide by the nonzero secant so the endpoint values become 0 and 1. The derivative of the Hermite cubic is then a quadratic whose coefficients depend only on the two normalized endpoint slopes. Substituting the harmonic-mean slopes and endpoint bounds places those normalized slopes in the monotonicity region derived by Fritsch and Carlson, so the quadratic is nonnegative on $[0,1]$ for increasing data and nonpositive for decreasing data. If adjacent secants change sign, the zero derivative joins the two monotone pieces without an extra extremum. The original paper contains the complete case-by-case quadratic inequalities.

F.3 Bezier and Hermite equivalence

Theorem (Cubic Representation Equivalence). A cubic Bezier curve with control points $P0=y_i$, $P1=y_i+h\, d_i/3$, $P2=y_{(i+1)}-h\, d_{(i+1)}/3$, $P3=y_{(i+1)}$ is exactly the cubic Hermite segment with endpoint values $y_i,y_{(i+1)}$ and slopes $d_i,d_{(i+1)}$.

Proof. Expand the four Bernstein polynomials in powers of t and collect the coefficients of y_i, $h\, d_i$, $y_{(i+1)}$, and $h\, d_{(i+1)}$. They are respectively $2t^3-3t^2+1$, t^3-2t^2+t, $-2t^3+3t^2$, and t^3-t^2, exactly the Hermite basis.

Appendix G. Bounded transforms and moment formulas

G.1 Logit enforces price bounds

Theorem (Logistic Bijection). The map $\text{logistic}(y)=1/(1+\exp(-y))$ sends every real y into $(0,1)$, is strictly increasing, and has inverse $\log(p/(1-p))$. Therefore $L+(U-L)\text{logistic}(y)$ always lies strictly between L and U.

Proof. The exponential is positive, so the denominator exceeds one and the fraction lies in $(0,1)$. Differentiation gives $\text{logistic}(y)[1-\text{logistic}(y)]>0$. Solving $p=1/(1+\exp(-y))$ for y gives the stated inverse. Affine rescaling maps $(0,1)$ to (L,U).

G.2 Expected clipped normal value

Theorem (Clipped Normal Mean). If $X\sim N(mu,s^2)$, then $E[\text{clip}(X,L,U)]=L+C(L)-C(U)$, where $C(K)=(mu-K)\Phi((mu-K)/s)+s\,\phi((mu-K)/s)$.

Proof. Pathwise, $\text{clip}(X,L,U)=L+(X-L)_+-(X-U)_+$. For a normal variable, substitute $X=mu+sZ$ in $E[(X-K)_+]$ and integrate over $Z>(K-mu)/s$. Splitting the integral into $(mu-K)$ times a tail probability plus s times the first normal tail moment gives $C(K)$.

G.3 Common random numbers

Theorem (Unbiased Common-Random-Number Labels). Using the same random draws to estimate prices at several states does not change the expectation of any individual state estimator. It changes only their cross-state covariance.

Proof. At each fixed state, the reused random vector has exactly the same marginal distribution as a freshly drawn vector, so its sample mean remains unbiased. Sharing draws correlates errors across states; for nearby smooth payoffs this usually makes the error itself vary smoothly, which helps interpolation.

Appendix H. Mathematical implementation tricks

This appendix turns the main implementation choices into mathematical claims. The statements are intentionally compact; each proof explains why the code transformation is valid, unbiased, stabilizing, or only a heuristic.

H.1 State enrichment is not importance sampling

Theorem (Two Different Measures). Let x be the pricing state and Z the simulated path randomness. Choosing more training states from a tail-focused state distribution nu_train changes where labels are requested. It does not change the conditional expectation inside any label. Importance sampling changes the simulation law for Z and must include a density-ratio weight.

```
State enrichment:
    x_i ~ nu_train,
    y_i = (1/N) sum_{n=1}^N h(x_i,Z_n),        Z_n ~ p_x.

Likelihood-ratio importance sampling:
    y_i = (1/N) sum_{n=1}^N h(x_i,Z_n) p_x(Z_n)/q_x(Z_n),
    Z_n ~ q_x.
```

Proof. For a fixed state x, the target label is E_p[h(x,Z)]. Drawing more states near a boundary only changes the empirical design measure used by the regression. It leaves each label estimator as an average under p_x. If paths are instead drawn from q_x, the expectation becomes E_q[h(x,Z)] unless the likelihood ratio p_x/q_x is multiplied pathwise. Therefore state enrichment improves coverage, while importance sampling improves rare-event label variance; they are related but not the same operation.

H.2 Sobol power-of-two batching

Theorem (Base-2 Digital-Net Balance). For a Sobol sequence in base 2, the first 2^m points form a digital net. Elementary binary boxes of compatible volume receive nearly uniform counts. Using powers of two therefore preserves the designed low-discrepancy balance better than stopping at an arbitrary path count.

```
Implementation rule:
    N_target -> N_sobol = 2^{ceil(log_2 N_target)}.

Training budget:
    N_state * N_sobol paths.
```

Proof. Sobol points are constructed so that the first 2^m points fill binary elementary intervals evenly according to the generator matrices. Truncating at a non-power-of-two count cuts through this balanced block. Rounding up keeps the full net and only increases work by less than a factor of two. This is a numerical-design theorem, not a probabilistic unbiasedness claim.

H.3 PCA compression of basket composition

Theorem (Best Linear k-Dimensional Reconstruction). Let R be the centered log-spot vector with covariance matrix Sigma. Among all rank-k orthogonal projections P, the projection onto the eigenvectors of Sigma with the k largest eigenvalues minimizes E[||R-PR||^2].

```
Sigma u_j = lambda_j u_j,    lambda_1 >= ... >= lambda_d.

P_k = U_k U_k',
score_j = R' u_j,       j=1,...,k.
```

Proof. For any orthogonal projection P with rank k, the retained variance is trace(P Sigma). Minimizing reconstruction error is equivalent to maximizing this retained variance because E[||R-PR||^2]=trace(Sigma)-trace(P Sigma). Rayleigh-Ritz gives the maximum trace as lambda_1+...+lambda_k, achieved by the span of the top eigenvectors. The basket Asian proxy uses the first four scores as a low-dimensional summary of relative basket composition.

H.4 Moment-matched lognormal baseline

Theorem (Two-Moment Lognormal Match). If a positive random variable A has mean m>0 and variance v>=0, the lognormal random variable exp(Y), Y~N(mu_L,s_L^2), with the same first two moments is obtained from s_L^2=log(1+v/m^2) and mu_L=log(m)-0.5 s_L^2.

```
s_L^2 = log(1 + v/m^2),
mu_L = log(m) - s_L^2/2.

V_LN = exp(-r tau) [ m Phi(d_1) - K Phi(d_2) ],
d_2 = (mu_L - log K)/s_L,
d_1 = d_2 + s_L.
```

Proof. For Y~N(mu_L,s_L^2), E[exp(Y)]=exp(mu_L+s_L^2/2) and Var(exp(Y))=exp(2 mu_L+s_L^2)(exp(s_L^2)-1). Setting the mean equal to m gives exp(mu_L+s_L^2/2)=m. Dividing the variance equation by m^2 gives v/m^2=exp(s_L^2)-1, hence the stated s_L^2 and mu_L. The option formula then follows by the same truncated-lognormal calculation used in Black-Scholes.

H.5 Ridge residual fitting

Theorem (Ridge Normal Equations). Given design matrix X, target y, and positive penalty lambda, the coefficient vector minimizing ||X beta-y||_2^2+lambda||beta||_2^2 is beta=(X'X+lambda I)^{-1}X'y. If the intercept is unpenalized, set the first diagonal penalty entry to zero.

```
min_beta  ||X beta - y||_2^2 + beta' Lambda beta

Gradient:
    2 X'(X beta-y) + 2 Lambda beta = 0

Solution:
    beta = (X'X + Lambda)^{-1} X'y.
```

Proof. The objective is a convex quadratic. Differentiating with respect to beta gives the displayed gradient. Setting it to zero yields the linear system. Adding lambda I makes the system better conditioned and shrinks unstable high-order basis coefficients. The code uses this for sparse Chebyshev and residual corrections.

H.6 Bounded log-factor correction

Theorem (Positive Multiplicative Correction). Let B(x)>=0 be a baseline price and eps>0. If a model fits a(x)=log((V(x)+eps)/(B(x)+eps)), then the reconstructed price V_hat(x)=(B(x)+eps) exp(a_hat(x))-eps is bounded below by -eps. Clipping at zero gives a nonnegative price.

```
a(x) = log( (V(x)+eps)/(B(x)+eps) )

V_hat(x) = max( (B(x)+eps) exp(a_hat(x)) - eps, 0 ).
```

Proof. The exponential is strictly positive, so (B+eps) exp(a_hat) is positive. Subtracting eps can only make the raw reconstruction larger than -eps. The final maximum with zero enforces the no-negative-option-price condition. Fitting a ratio rather than raw value lets the high-dimensional model learn relative bias around a strong moment baseline.

H.7 PCHIP residual calibration

Theorem (Conditional-Mean Residual Correction). Let M be a scalar summary feature and R=(V-g(X))/D(X) be a normalized training residual for a preliminary proxy g and positive scale D. Among all square-integrable functions c(M), the function c*(M)=E[R|M] minimizes E[(R-c(M))^2].

```
Preliminary proxy:     g(X)
Normalized residual:   R = (V - g(X)) / D(X)
Calibration function:  c*(m) = E[R | M=m]

Corrected proxy:
    V_cal(X) = max( g(X) + D(X) c_hat(M(X)), 0 ).
```

Proof. For any function c(M), condition on M. The conditional mean-square error is E[(R-c(M))^2 | M]. For a fixed value M=m this is minimized by the scalar mean E[R|M=m], because E[(R-a)^2|M=m]=Var(R|M=m)+(E[R|M=m]-a)^2. The code estimates c* by binning residuals in expected-average moneyness and fitting a PCHIP curve through the bin means. This is why the correction is a mathematical residual projection, not a hand-drawn patch.

H.8 Relative-error floor

Theorem (Stable Tail Error Denominator). For floor eta greater than zero, define e_rel=|V_hat-V|/max(|V|,eta). If |V| is at least eta, this is ordinary relative error. If |V| is below eta, then e_rel is bounded by |V_hat-V|/eta, so tiny option prices cannot create unbounded percentages from economically tiny absolute errors.

Proof. The first case follows from max(|V|,eta)=|V|. In the second case the denominator is eta, a fixed positive number. Thus the metric remains sensitive to absolute tail misses while avoiding division by a number arbitrarily close to zero.

H.9 Exact linear payoff wings

Theorem (Positive-Part Removal). If a call payoff can be written (Y-K)_+ and the state implies Y is at least K almost surely, then E[(Y-K)_+]=E[Y]-K. If the state implies Y is at most K, then E[(Y-K)_+]=0.

Proof. On the event Y is at least K, the positive part equals Y-K path by path. On the event Y is at most K, it equals zero path by path. Taking expectations preserves these identities. The scripts use this to avoid asking regressions to learn regions where the price is exactly linear or exactly zero.

H.10 Sparse Chebyshev basis selection

Theorem (Bounded Basis on a Scaled Domain). Chebyshev polynomials satisfy T_n(cos theta)=cos(n theta). Therefore |T_n(z)| is at most one for z in [-1,1]. Scaling inputs to this interval keeps each univariate basis column bounded before interactions are formed.

```
Scale feature:
    z_j = 2 (x_j - low_j)/(high_j-low_j) - 1.

Sparse design:
    1, T_1(z_j),...,T_p(z_j), selected z_i z_j, selected z_i z_j^2.
```

Proof. Every z in [-1,1] can be written z=cos theta for some theta in [0,pi]. Then T_n(z)=cos(n theta), whose absolute value is at most one. The code uses quantile scaling and clipping, so most training and validation features lie inside or near this bounded interval. Sparse term selection is a bias-variance tradeoff: it gives up full tensor expressiveness to avoid the combinatorial explosion of all interactions.

Appendix I. Primary references

These sources support the named theorems and numerical methods. The appendix statements are project-specific restatements, not copied quotations.

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Standard probability results (law of large numbers, central limit theorem, conditional expectation, and normal-density change of measure) are proved directly above in the finite-variance setting used by the simulations.